Our Reaction to Stress Explained: How to use ‘The Hand Model of the Brain’

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Dr. Daniel Siegel’s hand model of the brain allows us to picture our brain structure and understand why it’s difficult to control our reactions when we’re overwhelmed with strong emotions, especially stress.

Due to developmental factors, children are more susceptible to stress. The hand model of the brain helps them imagine what’s happening inside their brain when they get upset so that they can identify and deal with negative emotions more effectively.

First, let’s see what the hand model of the brain looks like.

As its name suggests, you need to use your hand for this. Your wrist is the **spinal cord** upon which the brain sits, your palm is the **inner brainstem**, and your thumb is your **amygdala** (or guard dog – see our blog on this here). If you place your thumb in the palm, you’ll form the **limbic system**. Your other fingers are your **cerebral cortex**, and the tips of your fingers are your **prefrontal cortex** (or wise owl). Close them on top of your palm and thumb, and there you have it – the hand model of the brain, divided into the upstairs brain (cerebral cortex) and downstairs brain (brainstem and limbic system).

So what exactly is happening in the ‘downstairs’ and ‘upstairs’ brain when we’re stressed?

Let’s start with the downstairs brain.

The brainstem regulates the basic functions of our body, like breathing and heart rate.

The limbic system is responsible for our emotions, memories, and activating instinctive survival reactions. It plays a crucial role in how we form relationships with other people and form a secure healthy attachment to them. More specifically, the **hippocampus** helps us form and save new memories, linking them to specific emotions. The **amygdala** is responsible for detecting danger and sends a signal to our autonomic nervous system to react accordingly.

Say, for instance, that you were stung by a bee once. The **hippocampus** has saved this memory, linking it to the fear you felt during the sting. Next time you see a bee, the **amygdala** (or guard dog) will identify the bee as a potential threat, and you’ll instinctively feel fear again.

At the same time, **cortisol**, a stress hormone, will be released. It will get your heart racing and your breathing to accelerate. (Remember how the brainstem controls both functions?) High levels of energy will be sent to your muscles so that you can get away from the threat as soon as possible. This is the fight or flight response.

Moving upstairs now, the **cortex** is our thinking brain. It’s the part of the brain that allows us to imagine, create, problem-solve, as well as grasp abstract notions and analyze them. The **prefrontal cortex,** in particular, helps us evaluate the situation detected by the amygdala (or guard dog) and control our emotional response before our feelings escalate to the point, we can’t control them.

Interesting to note is that the PFC does not finish developing and wiring until we are 25-30 years old!

What happens when we “flip our lid”?

We’ve seen how the brainstem and limbic system work together to release cortisol and prepare our body for the fight/flight/freeze response to protect us from danger. There’s nothing wrong with that. Besides, normally the prefrontal cortex kicks in and calms us down once we realize the threat is unreal or over.

However, this is not always the case. If there has not been a secure attachment prior to age 4 or early trauma/toxic stress experiences then this can affect the growth of the pathway link between guard dog (amygdala) and the wise owl (prefrontal cortex).

When we are in survival mode, we “flip our lid” and the downstairs brain takes over. Using the hand model of the brain, lift your fingers to release your fist – the connection between your upstairs and downstairs brain is lost. We can no longer effectively communicate with others, control our emotions, or respond to reasoning.

Note that we don’t have to be in real danger to flip our lid. Any stressful situation can cause our cortisol levels to rise and shut down our thinking brain. Since the prefrontal cortex isn’t fully developed until we’re 25, kids find it much harder to respond to stressful situations and regulate their emotions.

Explaining the hand model of the brain to a child

Use simpler terms to explain the hand model of the brain to a child.

Start with the palm that controls their basic body functions. The thumb represents big feelings like anger or stress and lives with the palm in the downstairs brain. The other fingers are the thinking brain, the upstairs brain. Show them how when they lift their fingers (“flip their lid”), their thinking brain is no longer connected to their feelings. That’s when they throw a tantrum or cry uncontrollably. Which is nothing unusual or to be ashamed of, but what if there was another way?

It’s as simple as curling their fingers over their thumb. Now their upstairs and downstairs brain are connected! They are embracing their feelings and listening to what their brain has to say about them. This is how we build healthier, more emotionally intelligent brains!

Dr. Daniel Siegel’s hand model of the brain is a great first step to helping your child get in touch with their emotions and explore healthy ways to release and regulate them. Along with your unconditional love and support, it can foster their emotional intelligence and help them build balanced relationships, so give it a try!

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2 thoughts on “Our Reaction to Stress Explained: How to use ‘The Hand Model of the Brain’”

Rick
January 18, 2021 at 8:52 am

I’m not that much of a online reader to be honest
but your blogs really nice, keep it up! I’ll go ahead
and bookmark your site to come back in the future.

Many thanks

Sian Chambers
October 6, 2024 at 10:57 am

Thank you very much :)

Comments are closed.

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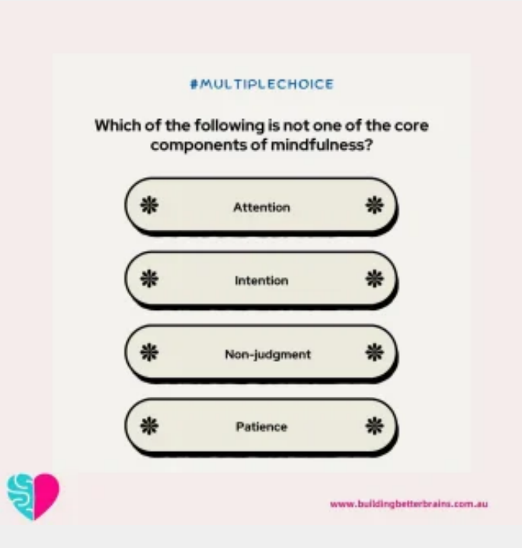
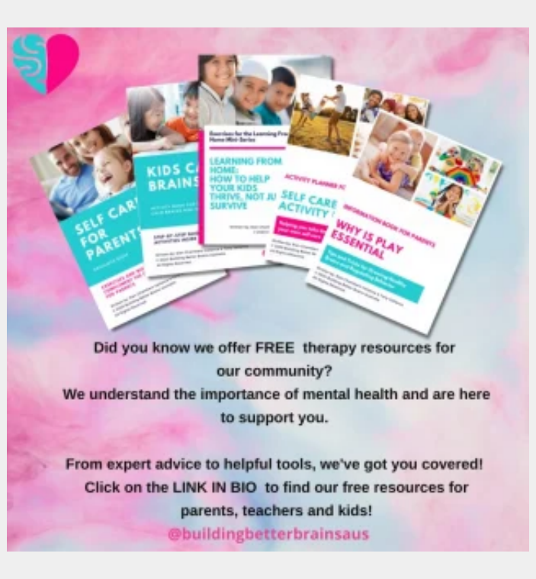

#MULTIPLECHOICE
Which of the following is not one of the core components of mindfulness?
Attention
Intention
Non-judgment
Patience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